Energy Consumption Optimization in Reverse Osmosis System (Daily Dataset)

Objective

To develop machine learning models that can:

- Predict daily energy consumption (energy_kwh) of an RO system.
- Identify key operational parameters influencing energy use.
- Recommend optimal settings to minimize energy consumption while maintaining system operation within feasible limits.

1. Dataset Overview

- Granularity: Daily records from the RO plant.
- Target Variable: energy_kwh
- Features: Influent and effluent flow rates, water quality parameters (e.g., pH, EC, TOC), pressure, temperature, and operational metrics.

2. Model Development and Evaluation

Three machine learning models were trained and compared for energy prediction accuracy.

Random Forest Regressor:

- R^2 Score: 0.915
- RMSE: 14,709.95 kWh
- Key Features: outflow_MG, input_pressure_psi, inflow_MG
- Quality: Accurate predictions with minimal error dispersion.

XGBoost Regressor:

- R^2 Score: 0.874
- RMSE: 17,927.13 kWh
- Key Features: inflow_MG, outflow_MG
- Quality: Slightly lower performance compared to Random Forest.

Support Vector Regressor (RBF Kernel):

- R^2 Score: 0.489

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- RMSE: 36,071.99 kWh
- Key Features: outflow_MG, input_pressure_psi, inflow_MG
- Quality: Underperformed due to limited non-linear learning capability for large-scale targets.

3. Optimization for Energy Minimization

Grid Search:

- Best Settings: inflow_MG=2500, outflow_MG=3100, rof_temp_F=70, rof_EC=1500, rof_pH=6.8, rop_pH=5.1, rop_EC=20
- Predicted Energy: 69,993 kWh
- Comment: Practical and realistic optimization result.

Optuna:

- Best Settings: Wide range across full feature space
- Predicted Energy: 994.19 kWh
- Comment: Unrealistically low inflow/output levels suggest theoretical result, not practically deployable.

Genetic Algorithm:

- Best Settings: inflow_MG=153.37, outflow_MG=113.52, rof_temp_F=66.24, rop_CO2=8859.41, ammonia=85.90
- Predicted Energy: 1018.54 kWh
- Comment: Similar to Optuna-very low energy but unrealistic operational values.

4. Insights & Recommendations

Key Drivers of Energy Consumption:

- High Impact: outflow_MG, input_pressure_psi, inflow_MG
- Moderate Impact: rof_temp_F, rof_EC, rop_pH
- Low Impact: ammonia, rop_CO2, rop_TOC

Best Performing Model:

- Random Forest Regressor showed the strongest performance and stability in prediction.

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Optimization Output:

- Grid Search produced the most feasible operational guidance.
- Optuna and GA highlighted minimum energy limits, but require value constraints for field application.

Optimization Methods for RO Energy Minimization

To identify the optimal combination of RO system operating parameters that minimizes energy consumption (`energy_kwh`) using trained predictive models (Random Forest, etc.).

Why Optimization is Needed

Predictive models alone estimate energy usage under known conditions but don't suggest how to change those inputs to reduce energy. Optimization techniques help us:

- Search through high-dimensional feature space.
- Respect physical/operational bounds.
- Find feature combinations that minimize energy, as predicted by the model.

Selected Optimizers

This study explored three optimizers: Grid Search, Optuna, and Genetic Algorithm (GA). Each has different mechanisms, advantages, and trade-offs suited to different aspects of this problem.

1. Grid Search

What it is:

Grid Search is a brute-force method that evaluates all combinations of input values across predefined ranges.

How it works:

- Define discrete sets of values for each variable (e.g., inflow_MG from 2500 to 3500).
- Evaluate all combinations using the trained model.
- Select the configuration that yields the lowest predicted energy.

Why it fits this use case:

- Deterministic and exhaustive.

- Suitable when feature ranges are well-understood.
- Produces interpretable, safe, and feasible results.

Limitations:
- Computationally expensive with many features or tight steps.
- Not effective for continuous, large-scale spaces.

2. Optuna (Bayesian Optimization)

What it is:

Optuna uses Bayesian optimization via Tree-structured Parzen Estimators to find optimal inputs intelligently.

How it works:
- Randomly evaluates a few configurations.
- Builds a probabilistic model of the energy surface.
- Prioritizes high-potential regions based on previous evaluations.

Why it fits this use case:
- Fast and adaptive to complex, continuous feature spaces.
- Doesn't require exhaustive enumeration.
- Handles non-linear interactions well.

Limitations:
- May find unrealistic solutions unless constrained.
- Results can vary with different trials or seeds.

3. Genetic Algorithm (GA)

What it is:

Genetic Algorithms mimic biological evolution using fitness selection, crossover, and mutation to optimize a problem.

How it works:

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- Start with a population of random input sets.
- Evaluate their performance (fitness).
- Combine and mutate top candidates to produce a new generation.
- Repeat until convergence.

Why it fits this use case:
- Effective for rugged, complex, non-convex energy landscapes.
- Handles feature interaction without assumptions.
- Explores wide search space.

Limitations:
- Computationally intensive.
- May need parameter tuning and feasibility filtering post-run.

Comparison Summary

Technique	Continuous Inputs	Scales Well	Deployment-Safe	Needs Constraints	Best For
Grid Search	No	No	Yes	No	Realistic, interpretable configs
Optuna	Yes	Yes	Sometimes	Yes	Intelligent fast exploration
Genetic Algorithm	Yes	Moderate	Sometimes	Yes	Complex nonlinear relationships

Each method complements the others and provides valuable insights under different priorities: safety, speed, or depth of search.